

$$W = K \left(\frac{x^2}{2} \right)$$

$$W = \frac{1}{2} K (x^2)_0^x$$

$$W = \frac{1}{2} K (x^2 - 0^2)$$

$$W = \frac{1}{2} K x^2$$

This work done is stored in the spring in the form of elastic potential energy.

$$\therefore U = \frac{1}{2} K x^2$$

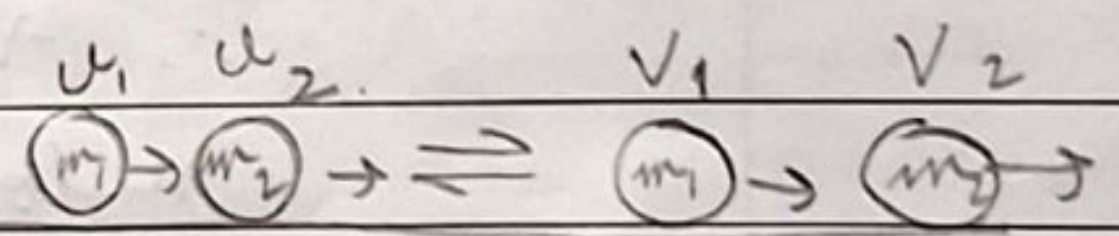
collisions.

collision is defined as an event in which two bodies exert force on each other and exchange linear momentum and kinetic energy with each other.

Coefficient of restitution (e)

For collision to take place.

$$u_1 > u_2$$



velocity difference before collision = $u_1 - u_2$ and is

known as velocity of approach.

$$v_2 > v_1$$

velocity difference after collision = $v_2 - v_1$ and

is known as velocity of separation.

$$e = \frac{\text{velocity of separation}}{\text{velocity of approach}}$$

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

Clearly, coefficient of restitution is unitless as well as dimensionless.

Types of collisions.

1) Elastic collision:

a) Linear momentum is conserved.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

b) Kinetic energy is conserved.

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

* For an elastic collision.

$$e = 1 = \frac{v_2 - v_1}{u_1 - u_2}$$

$$\therefore v_2 - v_1 = u_1 - u_2$$

velocity of separation = velocity of approach.

e.g., collision b/w subatomic particles.

2) Inelastic collision.

a) Linear momentum is conserved.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

b) Kinetic energy is not conserved.

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \neq \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

* for inelastic collision.

$$e < 1.$$

$$\frac{v_2 - v_1}{u_1 - u_2} < 1.$$

$$(v_2 - v_1) < (u_1 - u_2).$$

(velocity of approach > velocity of separation)

→ Almost all the collisions in day to day life are inelastic.

For an elastic collision.

$$e = \frac{v_2 - v_1}{u_1 - u_2} = 1.$$

Proof: Linear momentum is conserved.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2.$$

$$m_1 u_1 - m_1 v_1 = m_2 v_2 - m_2 u_2.$$

$$m_1 (u_1 - v_1) = m_2 (v_2 - u_2). \quad \text{--- (1)}$$

K.E is conserved.

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2.$$

$$m_1 u_1^2 - m_1 v_1^2 = m_2 v_2^2 - m_2 u_2^2.$$

$$m_1 (u_1^2 - v_1^2) = m_2 (v_2^2 - u_2^2). \quad \text{--- (2)}$$

Dividing eqn ② by eqn ①.

$$\frac{m_1(u_1^2 - v_1^2)}{m_1(u_1 - v_1)} = \frac{m_2(v_2^2 - u_2^2)}{m_2(v_2 - u_2)}$$

$$\frac{u_1^2 - v_1^2}{u_1 - v_1} = \frac{v_2^2 - u_2^2}{v_2 - u_2}$$

$$\frac{(u_1 + v_1)(u_1 - v_1)}{(u_1 - v_1)} = \frac{(v_2 + u_2)(v_2 - u_2)}{(v_2 - u_2)}$$

$$u_1 + v_1 = v_2 + u_2$$

$$u_1 - u_2 = v_2 - v_1$$

velocity of approach = velocity of separation.

$$\frac{v_2 - v_1}{u_1 - u_2} = 1 \Rightarrow e = 1$$

$$\frac{B - 9}{7 - 15} = 1$$

Q. Two bodies A and B moving with velocities 10 m/s and 3 m/s respectively collide with each other and after collision the velocity of body A is 8 m/s. Find the velocity of body B if the collision is elastic.

$$A = 10 \text{ m/s} = u_1$$

$$B = 3 \text{ m/s} = u_2$$

After collision
 $A = v_1 = 8 \text{ m/s}$

$$\text{Now, } u_1 - u_2 = v_2 - v_1$$

$$10 - 3 = v_2 - 8$$

$$7 = v_2 - 8$$

$$7 + 8 = v_2$$

$$v_2 = 15 \text{ m/s.}$$

Q. A force of ~~10 N~~ 12 N is applied on a body of mass 4 kg at rest and the body moves with an acceleration of

Q. A force of 12 N is applied on a body of mass 4 kg at rest. Find the kinetic energy of the body after travelling a distance of 5 m.

$$F = 12 \text{ N}$$

$$m = 4 \text{ kg}$$

$$u = 0$$

$$s = 5 \text{ m}$$

$$E = \frac{1}{2} m v^2$$

we know,

$$v^2 - u^2 = 2as$$

$$a = \frac{F}{m} = \frac{12}{4} = 3 \text{ m/s}^2$$

$$v^2 - 0^2 = 2 \times 3 \times 5$$

$$v^2 = 30$$

$$v = \sqrt{30} \text{ m/s} = 5.47$$



$$E = \frac{1}{2} m v^2$$

$$E = \frac{1}{2} \times 4 \times (5.47)^2$$

$$E = \frac{1}{2} \times 4 \times 30$$

$$E = 60 \text{ Jm}$$

Elastic collision in one dimension

Calculation of velocities (v_1 & v_2)

* Calculation of v_1

We have

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 (v_1 + u_1 - u_2)$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_1 + m_2 u_1 - m_2 u_2$$

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2 - m_2 u_1 + m_2 u_2$$

$$v_1 (m_1 + m_2) = (m_1 - m_2) u_1 + 2 m_2 u_2$$

$$v_1 = \frac{(m_1 - m_2) u_1}{(m_1 + m_2)} + \frac{2 m_2 u_2}{(m_1 + m_2)}$$

* Calculation of v_2

We have

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = m_1 (v_2 - u_1 + u_2) + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = m_1 v_2 - m_1 u_1 + m_1 u_2 + m_2 v_2$$

$$m_1 v_2 + m_2 v_2 = m_1 u_1 + m_2 u_2 + m_1 u_1 - m_1 u_2$$

$$(m_1 + m_2) v_2 = (m_2 - m_1) u_2 + 2 m_1 u_1$$

$$V_2 = \frac{(m_2 - m_1)}{(m_1 + m_2)} U_1 + \frac{2m_1 U_1}{(m_1 + m_2)}$$

Case: 1

If $m_1 = m_2 = m$ (say)

$$V_1 = 0 + \frac{2mU_2}{2m}$$

$$V_1 = U_2$$

$$\text{Also } V_2 = 0 + \frac{2mU_1}{2m}$$

$$V_2 = U_1$$

Case: 2

If we get body 2 at rest

$$U_2 = 0$$

$$V_1 = \frac{(m_1 - m_2)}{(m_1 + m_2)} U_1$$

$$V_2 = \frac{2m_1 U_1}{m_1 + m_2}$$

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Inelastic collision in one dimension.

calculation of velocities:

calculation of v_1 ,

we have,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 (eu_1 - eu_2 + v_1)$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 eu_1 - m_2 eu_2 + m_2 v_1$$

$$m_1 v_1 + m_2 v_1 = m_1 u_1 + m_2 u_2 - m_2 eu_1 + m_2 eu_2$$

$$v_1 (m_1 + m_2) = (m_1 - m_2 e) u_1 + m_2 u_2 (1 + e)$$

$$v_1 = \frac{(m_1 - m_2 e) u_1}{m_1 + m_2} + \frac{m_2 u_2 (1 + e)}{m_1 + m_2}$$

calculation of v_2 ,

we have,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = m_1 (v_2 - eu_1 + eu_2) + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = m_1 v_2 - m_1 eu_1 + m_1 eu_2 + m_2 v_2$$

$$m_1 v_2 + m_2 v_2 = m_1 u_1 + m_2 u_2 + m_1 eu_1 - m_1 eu_2$$

$$v_2 (m_1 + m_2) = (m_1 - m_2 e) u_1 + m_2 u_2 (1 + e)$$

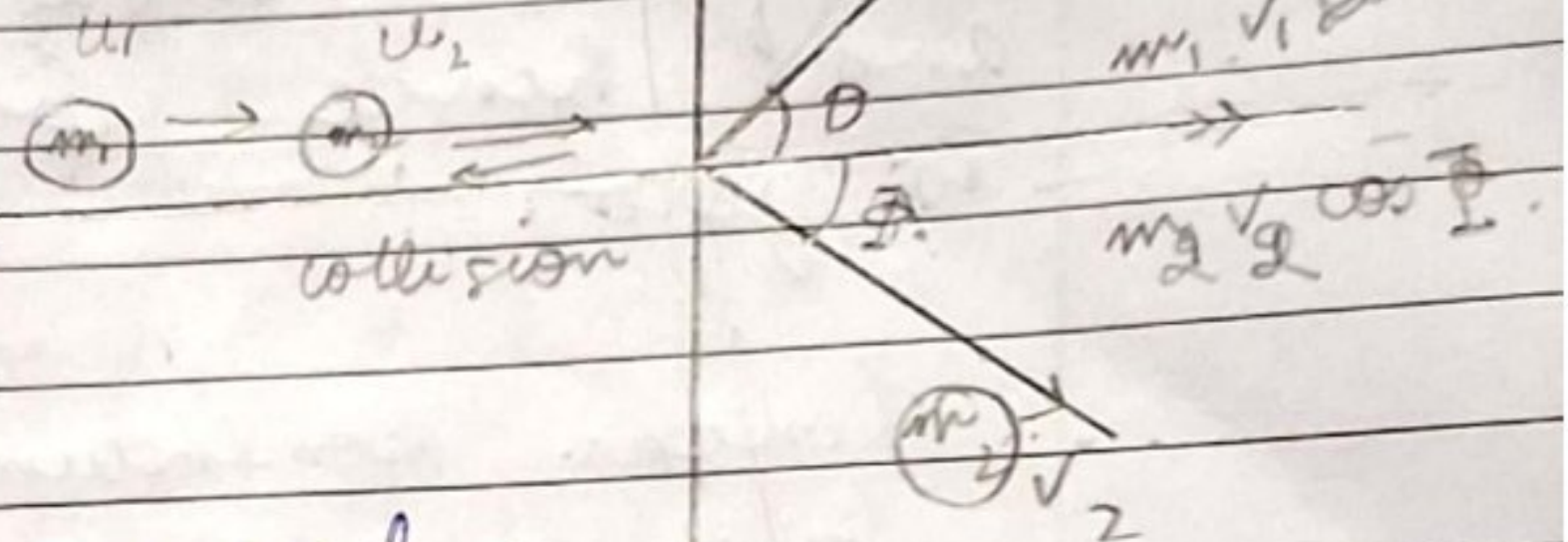
$$v_2 = \frac{(m_2 - m_1 e) u_2}{m_1 + m_2} + \frac{m_1 u_1 (1 + e)}{m_1 + m_2}$$

Elastic collision in two dimensions:

Kinetic energy is conserved.

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 =$$

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$



Linear momentum is conserved.

(hor)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi$$

(ver)

$$0 = m_1 v_1 \sin \theta - m_2 v_2 \sin \phi$$

$$m_1 v_1 \sin \theta = m_2 v_2 \sin \phi$$

Inelastic collision in two dimensions.

Kinetic energy is not conserved.

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \neq \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Linear momentum is conserved.

(hor)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi$$

(ver)

$$0 = m_1 v_1 \sin \theta - m_2 v_2 \sin \phi$$

$$m_1 v_1 \sin \theta = m_2 v_2 \sin \phi$$

Perfectly inelastic collision.

It is the collision in which the two bodies stick together and move with the same velocity.

Therefore, velocity of separation is zero and hence coefficient of restitution (e) is zero.

Linear momentum is conserved.

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

Kinetic energy is not conserved.

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \neq \frac{1}{2} (m_1 + m_2) v^2$$

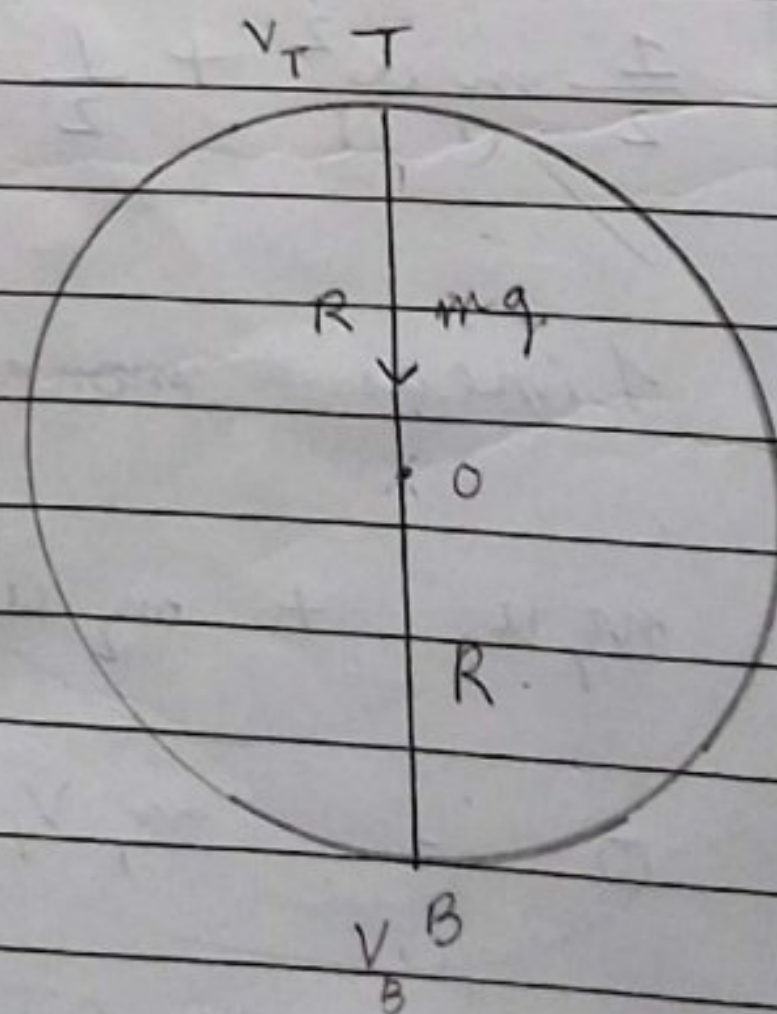
Motion in a vertical circle.

At top point (T)

$$mg = \frac{mv_T^2}{R}$$

$$v_T^2 = gR$$

$$v_T = \sqrt{gR}$$



At bottom point -

Total energy at bottom = Total energy at top.

$$\frac{1}{2} m v_B^2 + 0 = \frac{1}{2} m v_T^2 + mg(2R).$$

$$\frac{1}{2} m v_B^2 = \frac{1}{2} m g R + 2 m g R.$$

$$\frac{1}{2} m v_B^2 = m \left(\frac{1}{2} g R + 2 g R \right).$$

$$\frac{1}{2} v_B^2 = \frac{1}{2} g R + 2 g R \quad \frac{g R}{2} + \frac{2 g R}{2}$$

$$\frac{1}{2} v_B^2 = \frac{g R + 4 g R}{2} \quad \frac{2 g R}{2} + \frac{4 g R}{2}$$

$$v_B^2 = g R + 4 g R.$$

$$v_B = \sqrt{5 g R}.$$

Q A body of mass 5 kg, moving with a velocity of 8 m/s collides with a body of mass 2 kg at rest. Find the common velocity with which they move if the collision is perfectly inelastic.

$$V = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}.$$

$$m_1 = 5 \text{ kg}$$

$$m_2 = 2 \text{ kg}$$

$$u_1 = 8 \text{ m/s}$$

$$u_2 = 0 \text{ m/s}.$$



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$$V = \frac{5 \times 8 + 2 \times 0}{5 + 2}$$

$$V = \frac{40 + 0}{7} = \frac{40}{7} = 5.71 \text{ m/s.}$$